A Reconfigurable Built-In Self-Test Scheme for the Evaluation Circuits of Digital SRAM-IMC Architectures

Sunrui Zhang, Xiaole Cui, *Member, IEEE*, Feng Wei, and Xing Zhang

***Abstract*—Digital static random access memory-based in-memory computing (SRAM-IMC) is a promising computation paradigm to break the von-Neumann bottleneck. However, the IMC architectures also bring a series of challenges for testing, because of the circuit structures and operations that do not exist in the conventional memories. One of the challenges is the testing of evaluation circuits in the digital SRAM-IMC architectures, because the primary inputs (PIs) of the evaluation circuits cannot be directly accessed by the testers. Several test approaches such as the conventional logic built-in self-test (LBIST) modules, the indirect and the scan-chain-based test methods are proposed to address this issue. Nevertheless, these solutions suffer from the low test performance or the high area consumption. This work proposes a reconfigurable built-in self-test (BIST) scheme for the evaluation circuits. By reusing the IMC bitcells and operations, the proposed BIST scheme implements the separate pattern generation (PG) and response analysis (RA) processes. Furthermore, the diverse pattern generators, including the Fibonacci linear feedback shift register (LFSR) and weighted LFSR (WLFSR) with adjustable feedback polynomials and the cellular automata (CA), are realized to improve the test efficiency and fault coverage. The evaluation results show that the proposed BIST scheme has better test performance comparing with the indirect and the scan-chain-based test approaches. The proposed BIST scheme has comparable test performance, whereas it has much less area overhead comparing with the conventional LBIST schemes. Additionally, the proposed BIST scheme is testable and repairable.**

***Index Terms*—Built-in self-test (BIST), evaluation circuits, in-memory computing (IMC), reconfigurable, static random access memory (SRAM).**

I. Introduction

THE performance and power efficiency of the conventional von-Neumann computing architectures are limited by the relatively slow and high-power buses between processors and memories, although the modern semiconductor memories are with high I/O bandwidth [1], [2], [3]. The data-intensive applications, such as the graphic processing and the deep neural networks (DNNs), aggravate the problems of computation latency and energy consumption. The in-memory computing (IMC), as an emerging computation paradigm, is expected to be an effective memory-centric architecture to break through the von-Neumann bottleneck [4], [5], [6]. The digital static random access memory-based IMC (SRAM-IMC) architectures are of great concern, because the SRAM has advantages on data endurance and manufacturing maturity, and the digital IMC operations are robust against the process-voltage-temperature (PVT) variations [7], [8].

The digital SRAM-IMC architectures are different from the conventional SRAM in the aspect of circuit structure, as shown in Fig. 1. The original memory circuit modules, such as the memory bitcells and the address decoders, are modified in the digital SRAM-IMC architectures to realize the in-array logic and arithmetic computations [3], [9], [10]. Moreover, the additional evaluation circuits, such as the priority address encoder [11], the adder tree [12], [13], [14], [15] and the post

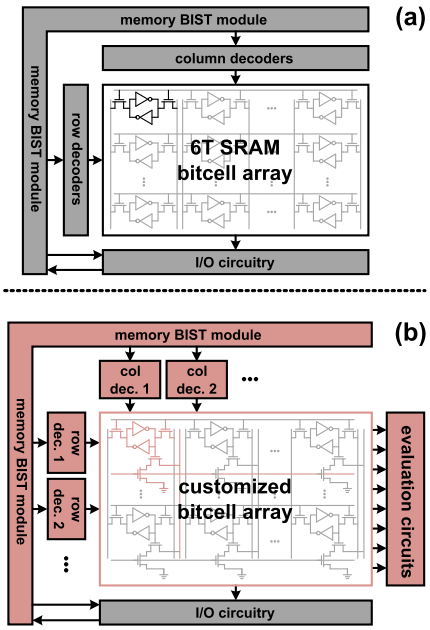


Fig. 1. Comparison between the digital SRAM-IMC architecture and the conventional SRAM. (a) Conventional SRAM. (b) Digital SRAM-IMC architecture. The different circuit blocks are highlighted.

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accumulator [16], [17], [18], are introduced to some digital SRAM-IMC designs to implement more complex Boolean and arithmetic computations. As an extension of the conventional memories, the IMC architectures are also vulnerable to the defects introduced by the manufacturing process. The unique circuits of IMC architectures cause some fault behaviors that differ from the known memory faults [19], [20], [21], [22], [23], [24]. In order to realize the practical and reliable digital SRAM-IMC architectures, it is crucial to develop the test solutions that aim to the special IMC features. The prior works on the digital SRAM-IMC architecture testing mainly focus on the fault models and test solutions for the bitcells [25], [26], [27], whereas the tests for the evaluation circuits are rarely discussed. In general, the commercial automatic test pattern generation (ATPG) tools can be utilized to generate the test patterns for the evaluation circuits, because most of them are regular combinational or sequential logic circuits. Nevertheless, the existing digital SRAM-IMC designs demonstrate two challenges of the evaluation circuit testing. First, the primary inputs (PIs) of circuits under test (CUT) only connect with the IMC bitcell array. Second, the D flip-flops (DFFs) do not always exist at the input ports of CUTs. Therefore, the external test patterns cannot be applied to the CUT directly.

Two test approaches, the indirect and the scan-chain-based test approaches, are proposed by Li [28] to address these challenges. The indirect test approach generates the expected test patterns from the bitcell array by executing a series of in-array logic operations. The scan-chain-based test approach inserts an additional chain of scan DFFs (SDFFs) between the bitcell array and the evaluation circuit. The external test patterns are shifted to the PIs of the CUT through the scan chain [29]. On the other hand, the conventional logic built-in self-test (LBIST) architectures, such as the built-in evaluation and self-test (BEST) architecture and the concurrent built-in logic block observer (CBILBO) [30], [31], [32], are also regarded as effective test solutions of the evaluation circuits.

However, the existing test methods have some drawbacks. The feasibility of the indirect test method depends on the in-array logic operations supported by the IMC architecture, and the method consumes substantial amount of clock cycles because of the iterative write and in-array computation operations. The scan-chain-based method is also inefficient, for the test patterns are shifted through a long scan chain. Although the test efficiency of the LBIST modules is higher than the former approaches, the independent registers and logic network consume extra area. Considering the high density of IMC bitcells, the area consumption of these peripheral circuits is quite unfriendly to the high-throughput IMC architectures.

The motivation of this work is developing an effective and efficient built-in self-test (BIST) solution for the evaluation circuits of digital SRAM-IMC architectures. The main contributions of the proposed BIST scheme are as follows.

1) The proposed BIST scheme combines the in-array and near-array computation techniques. The SRAM bitcells are reused to store the patterns and the signatures, meanwhile, the highly parallel IMC operations are utilized to substitute the functions of logic gates.

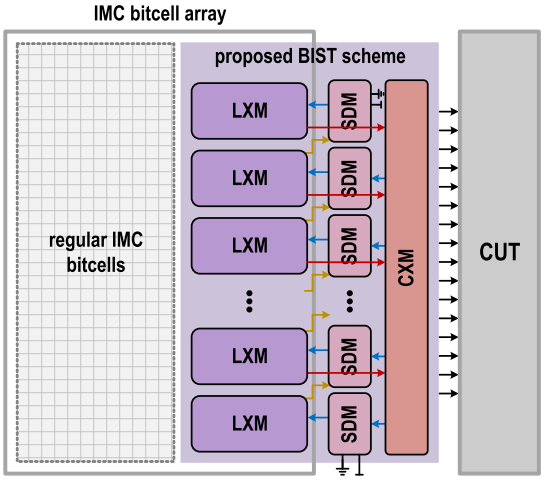


Fig. 2. Block diagram of the proposed BIST scheme for the evaluation circuits of digital SRAM-IMC architectures.

2) The proposed BIST scheme is reconfigurable. The stage number and the feedback polynomials of the linear feedback shift register (LFSR)/MISR can be adjusted without changing the hardware components. On the other hand, the proposed scheme supports diverse pattern generators, including the basic LFSR, the weighted LFSR (WLFSR), and the cellular-automata-based pseudorandom pattern generator (CA-based PRPG). The reconfigurability of the proposed scheme provides higher fault coverage, especially for the random-pattern resistant (RP-resistant) faults.
3) Comparing with the indirect and scan-chain-based test approaches, the proposed BIST scheme has better test performance. Comparing with the existing LBIST schemes, the proposed BIST scheme consumes less area. The proposed scheme makes a good balance between the test performance and area consumption, meanwhile maintains the high fault coverage.

The rest parts of the article are organized as follows. Section II presents the proposed BIST scheme. Section III illustrates the reconfigurability of the proposed BIST scheme. In Section IV, the evaluation results of the proposed scheme are presented and analyzed. The test and repair methods for the proposed scheme are discussed in Section V. Finally, the conclusion is drawn in Section VI.

II. Proposed BIST Scheme

A. Overall BIST Architecture

The general logic BIST process includes two main steps. First, the pattern generation (PG) step sends the test patterns to the CUT. Second, the response analysis (RA) step compacts the response data.

The Fibonacci LFSR and the MISR are realized by the proposed BIST scheme for PG and RA, respectively. Fig. 2 presents the overall block diagram of the proposed BIST scheme. The scheme contains a group of in-array local XOR modules (LXM), a group of near-array shift-drive modules (SDM), and a near-array multi-input chain XOR module (CXM). The customized IMC bitcells in the LXM group are

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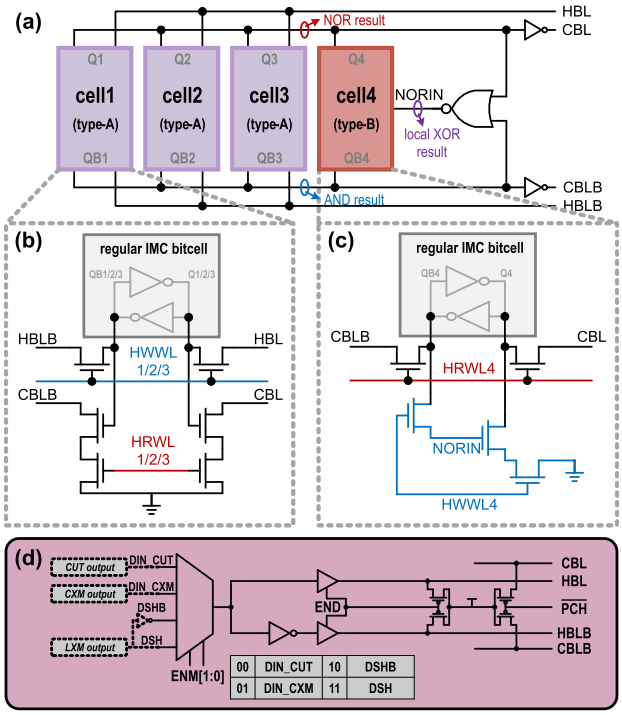


Fig. 3. Circuit structures of LXM and SDM. (a) Overall structure of LXM, which replaces the traditional DFF and implements the two-input local XOR function. The in-array NOR/AND results are generated on CBL/CBLB, respectively, and the local XOR result is generated by the NOR gate. (b) Structure of the type-A bitcells in LXM. (c) Structure of the type-B bitcell in LXM. The unique write port is marked in blue. (d) Structure of SDM, which facilitates the right shift and the horizontal memory operations.

used to replace the DFFs in the conventional LFSR/MISR. The SDM group is used to shift data between the LXMs or to facilitate the horizontal memory operations of the LXMs. The CXM is designed to implement the feedback XOR network of conventional LFSR/MISR.

The LXMs implement multiple two-input local XOR operations in parallel, and the results of the XOR operations are stored inside the module. The group of LXMs contains four columns of customized SRAM bitcells and a column of two-input NOR gates. Each LXM is embedded into a row of the IMC bitcells.

As shown in Fig. 3(a), an LXM includes three type-A bitcells, i.e., cell1, cell2, cell3, one type-B bitcell, i.e., cell4, and a two-input NOR gate. The circuit structure of the type-A bitcell is presented in Fig. 3(b). Based on the regular IMC bitcell structure, such as the 8T bitcell [3] or the 10T bitcell [10], each type-A bitcell includes the extra horizontal read and write ports. The horizontal decoupled read port is connected to a computation bitline pair CBL/CBLB, and it is controlled by the read wordlines HRWL1/2/3. The horizontal write port is connected to a write bitline pair HBL/HBLB, and it is controlled by the write wordlines HWWL1/2/3.

The circuit structure of cell4 is presented in Fig. 3(c). Apart from the regular IMC bitcell circuit, an extra read port and a unique write port are added to the type-B bitcell. The extra read port of cell4 is connected to CBL/CBLB and controlled by HRWL4. The unique horizontal write port, which is comprised by three transistors, is connected to the output of the two-input NOR gate and controlled by HWWL4. The NOR gate uses CBL/CBLB as its inputs. Additionally,

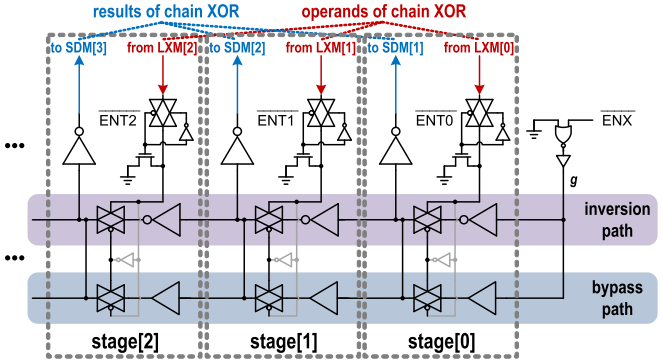


Fig. 4. Circuit structure of CXM, which implements the multi-input chain XOR operation. The red and blue arrows represent the inputs and outputs of the chain XOR operation, respectively.

two inverters are connected to CBL and CBLB, respectively, to output the data read out from the bitcells.

The circuit structure of SDM is shown in Fig. 3(d). Each SDM has four data input ports DIN_CUT, DIN_CXM, DSH, DSHB, one 2-bit port selection signal ENM[1:0], and a drive enable signal END. The DIN_CUT is connected to an output of CUT to receive the test response. The DIN_CXM is connected to an output of CXM to get the result of the chain XOR operation. For the rightmost SDM, i.e., the top SDM in Fig. 2, the DIN_CUT and DIN_CXM are connected to GND (logic "0") and V_{DD} (logic "1"), respectively. The DSH and DSHB are connected to the CBL and CBLB in the left adjacent row, respectively, to receive the shift data and its inversion. For the leftmost SDM, i.e., the bottom SDM in Fig. 2, the DSH and DSHB are connected to GND and V_{DD}, respectively.

The CXM implements the multi-input chain XOR operation. As shown in Fig. 4, the CXM contains multiple stages of inverters and transmission gates (TGs). Each stage includes two signal paths: the inversion path and the bypass path. The inversion path consists of an inverter and a TG connected in series, whereas the bypass path contains a buffer and a TG connected in series. The data passes through the two paths is determined by the enable signal $\overline{\text{ENX}}$. In each CXM stage, both the on-path TGs are controlled by the output of the corresponding LXM, and the LXM outputs are further controlled by an enable signal $\overline{\text{ENT}}$. The stage output is connected to the DIN_CXM port of the SDM in the left adjacent row.

Table I is the glossary of the primary signals in the proposed BIST scheme.

B. Memory Operations

The memory operations include the regular read/write operations and the unique horizontal read/write operations. On the one hand, the customized bitcells in LXMs are capable of implementing the regular read, write, and IMC operations, as the in-array regular IMC bitcells, because every customized bitcell contains the circuit structure of the regular IMC bitcell.

On the other hand, the bitcells in LXMs also implement the horizontal memory operations in parallel. In the horizontal read operation, the CBL/CBLB in each LXM is precharged, then the HRWL of bitcells is activated to read out the data. In the horizontal write operation, the data from CUT or CXM is

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TABLE I
GLOSSARY OF THE PRIMARY SIGNALS IN THE PROPOSED BIST SCHEME

Signal Names	Definitions
CBL/CBLB	The horizontal bitline pair for read and computation operations
HBL/HBLB	The horizontal bitline pair for write operations
HRWL*i*	The wordline of the horizontal read port of cell*i*
HWWL*i*	The wordline of the horizontal write port of cell*i*
DIN_CUT/DIN_CXM	The input port of SDM for receiving data from CUT/CXM
DSH/DSHB	The input port of SDM for receiving data from the CBL/CBLB output of LXM
END	The enable signal of SDM
ENM[1:0]	Two bits of port selection signals of SDM
$\overline{\text{PCH}}$	The enable signal of precharge circuit
$\overline{\text{ENT}i}$	The enable signal of the LXM output for CXM stage[*i*]
$\overline{\text{ENX}}$	The enable signal of CXM

written into the type-A bitcells of LXMs in parallel through the SDMs. For the data from CUT, the ENM[1:0] of the SDMs are set as "00" to select the DIN_CUT ports. For the data from CXM, the ENM[1:0] of the SDMs are set as "01" to select the DIN_CXM ports, and the $\overline{\text{ENT}}$s are set as "1." The $\overline{\text{ENX}}$ is set as "0" or "1" to send "0" or "1" to the LXMs, respectively. One exception is that the rightmost LXM gets the fixed "0" from DIN_CUT, and gets the fixed "1" from DIN_CXM.

C. Right Shift Operation

The yellow arrows in Fig. 2 illustrates the right shift operation. The data is read out from the source bitcell and written back to the destination bitcell in the right adjacent LXM. First, precharge the CBL/CBLB of the source LXM, then open the read port of the source bitcell. Meanwhile, activate the END, and configure ENM[1:0] as "11" or "10" to select the DSH or DSHB port. The data stored in the same bitcell column are right-shifted in parallel, because the horizontal read and write operations in each LXM are executed simultaneously. A fixed "0" is shifted into the leftmost LXM during the right shift operation.

D. Local XOR Operation

The operands of the two-input local XOR operation are stored in any two of the type-A bitcells in LXM. Assume the operands are stored at cell1 and cell2, respectively, the local XOR operation is implemented as follows. First, the computation bitlines CBL/CBLB are precharged to V_{DD}. Then, the HRWL1, HRWL2, and HWWL4 are activated to open the decoupled read ports of cell1 and cell2, and the unique write port of cell4 at the same time. If either cell1 or cell2 stores "1," i.e., Q1 or Q2 stores "1," the CBL is discharged. Similarly, if either cell1 or cell2 stores "0," the CBLB is discharged. Thereby, the NOR and AND results of the operands at Q1 and Q2 are obtained at CBL and CBLB, respectively, and the local XOR result is written to Q4 through the NOR gate, as presented in Fig. 3(a).

E. Chain XOR Operation

The multi-input chain XOR operation is realized based on the following Boolean equations:

$$g \oplus 1 = \bar{g} \tag{1}$$

$$g \oplus 0 = g \tag{2}$$

where "⊕" denotes the XOR operator. The chain XOR operation is implemented as follows. Set $\overline{\text{ENX}}$ as "0," then a logic signal $g = 1$ is sent to the inversion path and the bypass path. The operands of the chain XOR operation are read out from LXMs to control TGs in the CXM stages. Set $\overline{\text{ENT}}$s of the CXM stages as "0." If the operand "1" is loaded into a CXM stage, the inversion path of the stage is enabled, whereas the bypass path is cut off. In this case, the logic signal g is inverted, which is equivalent to the XOR operation "g⊕1." On the contrary, the bypass path is enabled if the operand "0" is loaded. In this case, the logic signal from the former stage passes by, which represents the XOR operation "g⊕0." Consequently, the final logic function F_{CXM} of an n-stage CXM circuit is equivalent to an n-input XOR operation, as shown in the following equations:

$$\begin{aligned} F_{\text{CXM}} &= \overline{1 \oplus (x_0 \oplus x_1 \cdots \oplus x_{n-2} \oplus x_{n-1})} \\ &= x_0 \oplus x_1 \cdots \oplus x_{n-2} \oplus x_{n-1} \end{aligned} \tag{3}$$

where x denotes an operand of chain XOR operation. In addition, if $\overline{\text{ENX}}$ is set as "1" during the operation, the corresponding logic function is the inversion of F_{CXM}, which is noted as "chain $\overline{\text{XOR}}$." It is equivalent to the n-input XNOR operation when n is even. The results of chain XOR/$\overline{\text{XOR}}$ operations are written back to the bitcell array by the last CXM stage participating in the computation.

F. Basic BIST Workflow

Comparing with the conventional n-stage Fibonacci LFSR and MISR, the n DFFs in the LFSR/MISR are replaced by n LXMs. The feedback XOR gates and the tap coefficients are replaced by the CXM, and the shifting operations between the DFFs are implemented by the SDMs and a temporary LXM.

To implement the BIST process, the proposed BIST scheme works as n-stage Fibonacci LFSR and n-stage MISR in alternate. The workflow of the first BIST iteration is illustrated as the following steps.

1) *LFSR Initialization:* An n-bit LFSR seed is written to the n cell1s of LXM[0], LXM[1], …, LXM[n-1] by two horizontal write operations. All the "0" bits and all the "1" bits of the LFSR seed are written separately through the DIN_CXM ports.
2) *MISR Initialization:* An n-bit MISR seed is written in the n bitcells from cell2[0] to cell2[n-1]. Similar to the LFSR initialization, the MISR initialization also requires two horizontal write operations.
3) *LFSR Output and Chain XOR:* The n-bit data stored in the cell1s is read out and sent to the CUT as a test pattern. Meanwhile, the test pattern is input to CXM as the operands of the multi-input chain XOR operation. The $\overline{\text{ENX}}$ of CXM and the $\overline{\text{ENT}}$s of the CXM stages with tap coefficient "1" are activated. Subsequently, the

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Fig. 5. Workflows of the proposed BIST scheme. The temporary LXM[n] and SDM[n] are marked by the dashed lines. (a) LFSR initialization. (b) MISR initialization. (c) LFSR output and chain XOR. (d) Response capture. (e) MISR chain XOR. (f) MISR local XOR. (g) LFSR right shifting. (h) MISR right shifting.

feedback XOR result of LFSR is stored in the cell1 of a temporary LXM, i.e., LXM[n].

4) *Response Capture:* The n-bit response of CUT is written to cell3[1], cell3[2], …, cell3[n] in parallel through the DIN_CUT ports of SDM[1], SDM[2], …, SDM[n].
5) *MISR Chain XOR:* The n-input chain XOR result is obtained by CXM, and it is temporarily stored in cell2[n]. This step is similar to step (3).
6) *MISR Local XOR:* The signature stored in cell2s and the test response stored in cell3s are compacted by the parallel local XOR operations executed in LXM[1], LXM[2], …, LXM[n]. The compacted signature is stored in the cell4s of these n LXMs.
7) *LFSR Right Shifting:* The data stored in cell1[n], cell1[n-1], …, cell1[1] are shifted to cell2[n-1], cell2[n-2], …, cell2[0] through the DSH ports of SDM[n-1], SDM[n-2], …, SDM[0].
8) *MISR Right Shifting:* The n bits of the compacted signature stored in cell4[n], cell4[n-1], …, cell4[1] are shifted to cell1[n-1], cell1[n-2], …, cell1[0]. The data in the cell1s is safely overwritten, because all the related operations have been finished. Note that the data is shifted through the DSHB ports rather than the DSH ports, because the data read from the type-B bitcells are not inverted.

The eight steps in the aforementioned workflow are shown in Fig. 5(a)–(h), respectively. In each subsequent BIST iteration, the proposed BIST scheme executes steps (3)–(8) to generate the test pattern and compact the test response. As presented in Fig. 6, a data mapping strategy is developed to avoid storing the data into the conflicting locations. Since the PG and the signature analysis are separated, the proposed BIST scheme can be used for the exhaustive or pseudo-exhaustive testing.

III. Reconfigurability of the Proposed BIST Scheme

The digital SRAM-IMC designs in previous research contain various evaluation circuits. The test efficiency of the BIST scheme with the specific pattern generator and response analyzer varies for the different evaluation CUT. Moreover, the emerging adaptive data-driven test methods also require real-time dynamic adjustment of test patterns according to the features and distribution of test data to reduce test cost [33], [34]. Therefore, the reconfigurability of the BIST schemes are critical to meet diverse test needs.

In the proposed BIST scheme, the LXM and CXM are capable of implementing multiple kinds of computation operations. In addition to the local XOR operation, the LXMs can also execute the in-array AND and NOR operations with the

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		cell1 column	cell2 column	cell3 column	cell4 column
1st BIST workflow	*LFSR Initialization*	LFSR seed (1st test patt.)	no data	no data	no data
	MISR Initialization	1st test patt.	MISR seed	no data	no data
	LFSR Output & Chain XOR	2nd test patt.	MISR seed	no data	no data
	Response Capture	2nd test patt.	MISR seed	1st test resp.	no data
	MISR Chain XOR	2nd test patt.	MISR seed	1st test resp.	no data
	MISR Local XOR	2nd test patt.	MISR seed	1st test resp.	1st signature
	LFSR Right Shifting	2nd test patt.	2nd test patt.	1st test resp.	1st signature
	MISR Right Shifting	1st signature	2nd test patt.	1st test resp.	1st signature
2nd BIST workflow	*LFSR Output & Chain XOR*	1st signature	3rd test patt.	1st test resp.	1st signature
	Response Capture	1st signature	3rd test patt.	2nd test resp.	1st signature
	MISR Chain XOR	1st signature	3rd test patt.	2nd test resp.	1st signature
	MISR Local XOR	1st signature	3rd test patt.	2nd test resp.	2nd signature
	LFSR Right Shifting	1st signature	3rd test patt.	3rd test patt.	2nd signature
	MISR Right Shifting	2nd signature	3rd test patt.	3rd test patt.	2nd signature
3rd BIST workflow	*LFSR Output & Chain XOR*	2nd signature	3rd test patt.	4th test patt.	2nd signature
	Response Capture	2nd signature	3rd test resp.	4th test patt.	2nd signature
	...	...	...	...	...

Fig. 6. Data map of the bitcell columns in LXMs during the test process. The data related to LFSR and MISR workflows are marked in red and blue, respectively.

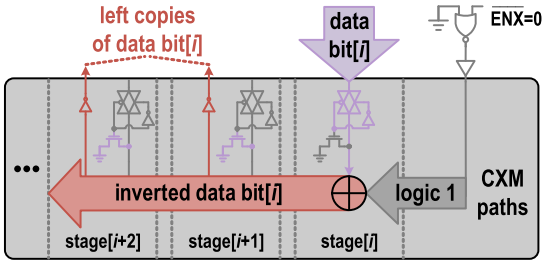


Fig. 7. Single-bit left copy operation implemented by CXM. The purple and the red arrows represent the source and the destination of the copy operation, respectively.

operands stored in the type-A bitcells. Besides the chain XOR operation, the CXM is able to implement the single-bit left copy operation. As shown in Fig. 7, the data bit[i] read from a bitcell can be written back to the other bitcells in any of the left LXMs, if the $\overline{\text{ENX}}$ is set as "0" and all $\overline{\text{ENT}}$s except the $\overline{\text{ENT}}[i]$ are set as "1."

Taking advantage of the versatility of LXMs and CXM, the proposed BIST scheme can be reconfigured to realize multiple extra functions to improve the fault coverage and the test efficiency. A summary of the extra functions is illustrated in Table II.

A. Adjustable Stage Number and Feedback XOR Network

Adjusting the stage number of LFSR/MISR improves the fault coverage of BIST schemes. For instance, the MISR with more stages significantly decreases the aliasing probability of RA, thereby the chance of test escape is smaller [35]. Since each LXM of the proposed BIST scheme represents a stage of the LFSR/MISR, the stage number of the LFSR/MISR can be adjusted by simply using more or less LXMs and CXM stages in the BIST workflow.

Meanwhile, the feedback XOR network of the proposed LFSR/MISR can also be reconfigured. By activating or deactivating different $\overline{\text{ENT}}$ signals of CXM stages, different data

TABLE II
EXTRA FUNCTIONS REALIZED BY THE PROPOSED BIST SCHEME

Functional Extensions	Improvements
LFSR/MISR stage number adjustment	Increasing fault coverage by lowering aliasing probability
Feedback XOR network adjustment	Generating deterministic patterns by enabling LFSR reseeding
WLFSR & CA-based PRPG	Decreasing test length and increasing fault coverage for the random-pattern resistant faults

stored in the LFSR/MISR can be selected to participate in the feedback XOR functions. The primitive polynomial for the LFSR/MISR with a different number of stages are realized by the reconfigurable feedback network. Furthermore, the changeable feedback polynomials also enable the LFSR reseeding, by which the deterministic test patterns are generated. Previous work confirmed that the mixed pseudorandom and deterministic patterns achieve higher fault coverage [36]. For instance, the fault simulation results on benchmark circuits C2670 and C7552 in ISCAS85 reveals that, with a test length of 1000, the fault coverages of pseudorandom patterns are 87.9% and 92.7%, respectively. However, the fault coverages of the mixed patterns hit 99.5% and 100%, respectively [36].

B. Diverse PG Techniques

Besides the Fibonacci LFSR, the proposed scheme is able to implement multiple kinds of pattern generators, including the WLFSR and the CA-based PRPG. The WLFSR generates test patterns with special proportions of "0" to "1," while the CA-based PRPG provides the test patterns with better randomness. Both the WLFSR and the CA-based PRPG shorten the test length and enhance the coverage for the RP-resistant faults in CUTs. For the WLFSR, the fault simulation results on C2670 and C7552 circuits indicate that the weighted random patterns have 11.7% and 4.1% higher fault coverages, respectively, than the equiprobable random patterns [37]. For the CA-based PRPG, the simulation results also present that, on C2670 and C7552 circuits, the BIST with CA-based PRPGs has 17.1% and 59.9% less undetected faults than the BIST with regular LFSR [38].

The WLFSR in the proposed BIST scheme is realized by ANDing and ORing the test patterns generated by more than one independent LFSRs. Fig. 8(a) presents a WLFSR with a weight factor of 3, which means that the WLFSR generates test patterns with 3/4 and 1/4 probabilities of being "1." The three columns of type-A bitcells in LXMs are utilized to implement two independent LFSR workflows. Each time after the patterns are generated, the data bits of the two patterns are ORed and ANDed in parallel. The OR and AND results of the two test patterns correspond to the weighted test pattern with 3/4 and 1/4 probability of being "1," respectively.

Furthermore, if each LXM in the proposed BIST scheme is extended to contain m type-A bitcells, up to $m-1$ independent LFSRs can be realized, thereby the WLFSR can generate test patterns with a maximum weight factor of $2^{m-1}-1$. The weighted bits obtained from the WLFSR are hardly correlated,

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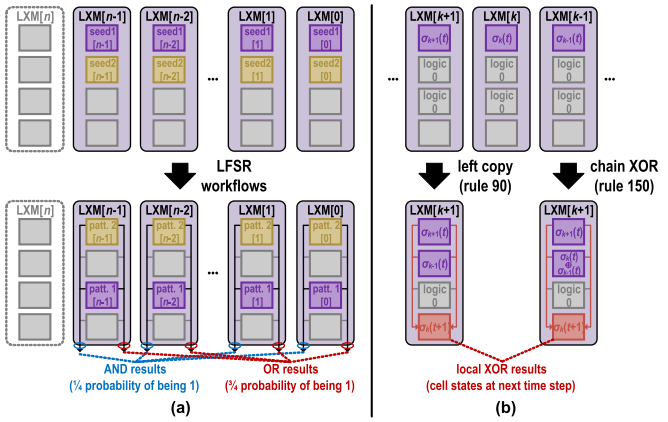


Fig. 8. WLFSR and the CA-based PRPG realized by the proposed BIST scheme. (a) WLFSR. The weighted bits are generated on the CBL/CBLB of each LXM. (b) CA-based PRPG. The next state of CA is generated by the local XOR operation.

because the original test patterns used for generating the weighted test patterns come from the independent LFSRs with different seeds.

The CA-based PRPG circuit in the proposed BIST scheme exploits the chain XOR and the left copy functions of CXM. Fig. 8(b) presents a 1-D CA using alternate state transition rules 90 and 150. The rules 90 and 150 are defined in the following equations [32], [38]:

$$\sigma_k(t+1)_{90} = \sigma_{k-1}(t) \oplus \sigma_{k+1}(t) \tag{4}$$

$$\sigma_k(t+1)_{150} = \sigma_{k-1}(t) \oplus \sigma_k(t) \oplus \sigma_{k+1}(t) \tag{5}$$

where $\sigma_k(t)$ denotes the state of cell$_k$ in CA at time t. The n-bit data stored in LXM[0], LXM[1], …, LXM[n-1] is regarded as the current state of CA. In the case of rule 90, $\sigma_{k-1}(t)$ is left copied from LXM[k-1] to LXM[k+1] directly. In the case of rule 150, $\sigma_{k-1}(t)$ and $\sigma_k(t)$ are first chain XORed by CXM, and the result is written to LXM[k+1]. The left copy or the chain XOR operation is executed for each cell in CA, respectively, depending on the corresponding state transition rules. Afterward, the parallel local XOR operations are implemented in all n LXMs to generate the state $\sigma(t+1)$ for each cell in CA. Finally, the n-bit state obtained in the type-B bitcells is right-shifted to the original LXMs and output as the generated test pattern. Note that all the type-A bitcells are initialized to "0" to guarantee that the state transition rules are correctly implemented in each LXM.

IV. Evaluation Results

A. Effectiveness of the Proposed BIST Scheme

To validate the function, the proposed BIST scheme is set to implement BIST for a 2-bit multiplier including a 2×2 Wallace tree and a carry propagating adder, which are the typical evaluation circuits of the digital SRAM-IMC architectures. The multiplier has two input ports for each 2-bit operand, and four output ports for the result of multiplication. In the BIST process, the workflows of 4-bit Fibonacci LFSR and 4-bit MISR are executed alternately. The seed of the LFSR and the MISR are set as "1111" and "0101," respectively, so the subsequent test patterns generated by the LFSR are

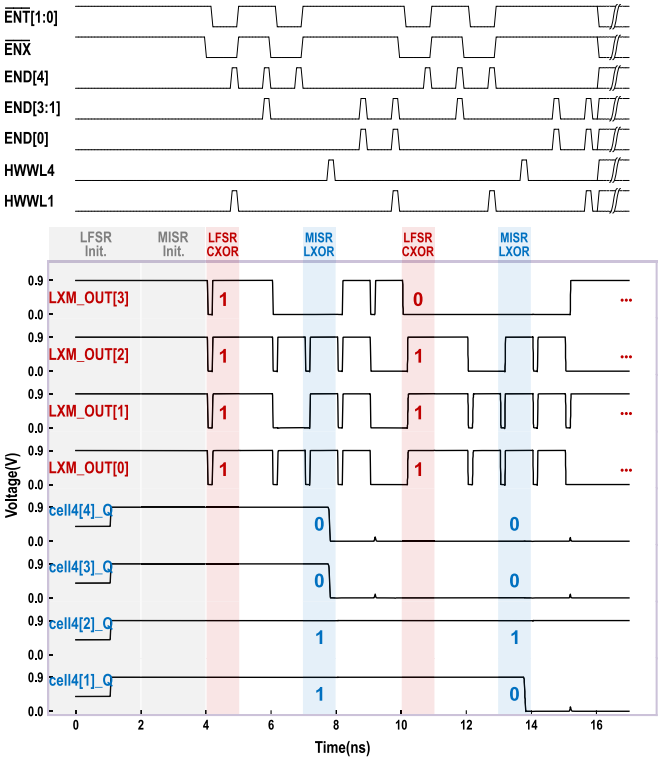


Fig. 9. Timing diagram and simulation waveforms of the first two iterations of the BIST process for the 2-bit multiplier. The test patterns and the signatures are marked in red and blue, respectively.

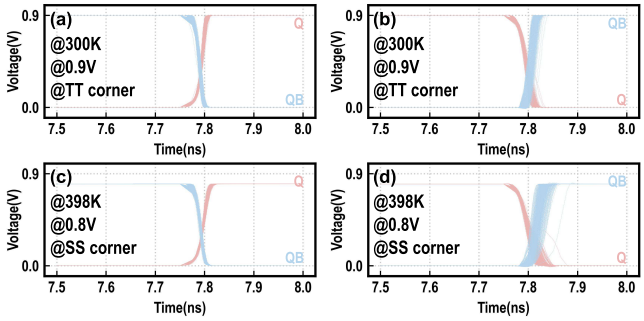


Fig. 10. 1000-point Monte-Carlo simulation results of the storage node voltages during the local XOR operation. The results are "1" in (a) and (c) under typical and slow PVT conditions, respectively. The results are "0" in (b) and (d) under typical and slow PVT conditions, respectively.

"0111→0011, …." Assume the multiplier is fault-free, the corresponding test responses are "1001→0011→0000, …," and the compacted signatures are "0011→0010→1001, …." The workflows are simulated with TSMC 28 nm HPC technology files. Fig. 9 presents the timing diagram and the waveforms of the LFSR and MISR workflows in the first two iterations. The waveforms show that the test patterns and the signatures obtained from the proposed BIST scheme are consistent with the expected outputs of the standard Fibonacci LFSR and MISR.

In order to evaluate the robustness of the unique horizontal write ports of the type-B bitcells, the local XOR operations in the BIST process are simulated. Fig. 10 presents the storage node voltages of the type-B bitcells under the 1000-point Monte-Carlo simulations during the local XOR operations. Different PVT conditions are also considered. The typical

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PVT conditions are the nominal power supply of 0.9 V, 300K temperature, and TT process corner. The slow PVT conditions are 10% lower power supply voltage, 398K temperature, and SS process corner. The waveforms verify the robustness of the unique horizontal write operations against the PVT variations, even though the write operation is single-ended. The robustness comes from the structure of the unique write port. It is able to avoid the voltage loss during writing either "1" or "0," and the noise on CBL/CBLB is filtered by the NOR gate.

The proposed BIST scheme can be physically implemented along with the digital SRAM-IMC designs using current technologies. The customized bitcells in the LXMs are designed based on the regular IMC bitcells, so the width of their layouts can be easily adjusted to meet that of the other bitcells. Moreover, the impact on array density is also acceptable. The customized bitcells only occupy a small proportion of total bitcells in the array, and they are realized with design rules for memories, which allow denser placement and routing. On the other hand, the near-array parts of the proposed BIST scheme are typical logic circuits that consist of standard digital gate cells. As a result, these circuits can be deployed as the other peripheral circuits of the bitcell array.

B. Test Performance and Area Overhead

To evaluate the overall performance of the test process, the total clock cycle counts of a single test iteration, i.e., the total number of cycles from the start point of one PG to the endpoint of the corresponding response compaction, is used as the metric.

The proposed BIST scheme in Fibonacci LFSR mode takes two clock cycles to generate a test pattern. One cycle is for the chain XOR, and the other cycle is for the right shifting. In WLFSR mode, the cycle count for PG depends on the weight factor. For the WLFSR with a weight factor of f, $\log_2(f+1)$ independent LFSRs are included. Each LFSR needs two cycles to obtain an original test pattern, and one extra cycle is needed to output the weighted test patterns, so the total cycle count is $2\log_2(f+1)+1$. The cycle counts of both LFSR and WLFSR modes are unrelated to the number of PIs of CUT. In CA-based PRPG mode, the cycle consumption increases linearly with the amount of cells in the CA. For the CA-based PRPG with c cells, the cycle count to generate each test pattern is c +3. The first c cycles are used to configure the state transition rules for each cell in CA, and the last three cycles are consumed by the local XOR, the right shifting, and the output operations. Finally, the proposed BIST scheme consumes four cycles to compact a test response in MISR mode.

Fig. 11 presents the cycle counts of a single BIST iteration in different PG modes. To implement a BIST iteration for the CUT with 128 PIs, the proposed BIST scheme consumes 6, 17, and 135 cycles with Fibonacci LFSR, WLFSR with a weight factor of 63, and CA-based PRPG, respectively.

The area overhead of the proposed BIST scheme is characterized by the extra amount of transistors. For the SDMs and CXM stages, all the transistors are taken into account. The transistor counts of the single SDM and CXM stages are 30 and 19, respectively. For the LXMs, only the transistors

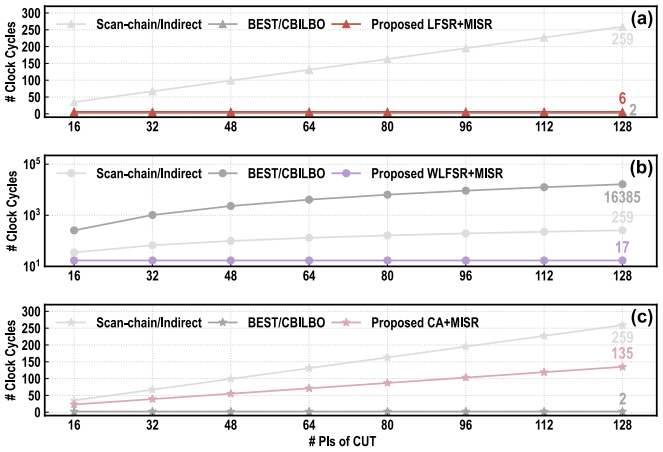


Fig. 11. Comparison of single test iteration performance in different PG modes. (a) Fibonacci LFSR. (b) WLFSR with a weight factor of 63. (c) CA-based PRPG.

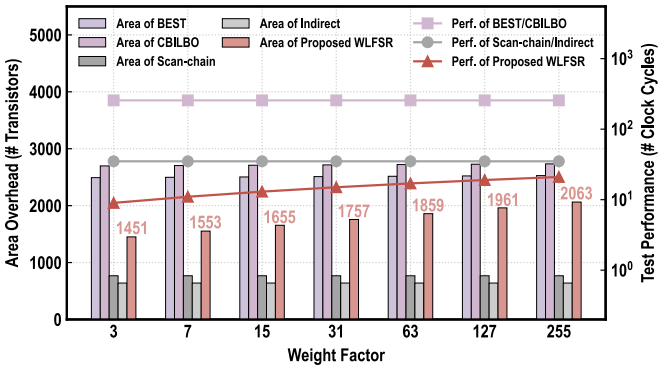


Fig. 12. Comparison of area overhead and test performance in the WLFSR mode with different weight factors. The stage of the WLFSR is set as 16.

comprise the horizontal read and write ports of the bitcells are taken into account. The reason is that the bitcells can work as the regular IMC bitcells when the test function is not activated, the actual area overhead of LXMs only comes from the horizontal read/write ports used for the BIST process. The area overhead of a single LXM depends on the maximum weight factor supported by the proposed scheme. Since the WLFSR mode requires multiple independent LFSRs, the LXMs include more bitcells to store the test patterns. To implement a WLFSR with maximum weight factor f, the number of type-A bitcells included by each LXM is $\log_2(f+1)+1$. In addition, one type-A and one type-B bitcells are required by the MISR mode. Each type-A bitcell contains six extra transistors, while the type-B bitcell and the rest parts of LXM consume 13 transistors. Consequently, the total area overhead of a single LXM is $6(\log_2(f+1)+2)+13$.

Table III illustrates the area overhead of the proposed BIST scheme with different PI amounts of CUT. For the PI number of 16 and 128, the proposed scheme with a maximum weight factor of 63 consumes 1859 and 14 179 transistors, respectively. Fig. 12 shows the area overhead of the proposed 16-stage WLFSRs with different weight factors. The proposed WLFSR consumes 1451 and 2063 transistors with weight factors 3 and 255, respectively.

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TABLE III
AREA OVERHEAD (# TRANSISTORS) OF DIFFERENT TEST SOLUTIONS

# PIs*	weight factor=63				
	Indirect	Scan-chain	BEST	CBILBO	**Proposed**
16	640	1280	2516	2722	**1859**
32	1280	2560	4932	5330	**3619**
48	1920	3840	7348	7938	**5379**
64	2560	5120	9764	10546	**7139**
80	3200	6400	12180	13154	**8899**
96	3840	7680	14596	15762	**10659**
112	4480	8960	17012	18370	**12419**
128	5120	10240	19428	20978	**14179**

* Assume # PIs equals to # POs.

C. Reconfigurability Cost

Fig. 13 illustrates the state transition graph of the proposed BIST scheme in different modes. Each of the PG workflow is bound with a RA workflow, as a complete BIST iteration. Every BIST iteration is designed to start from the IDLE state. The PG modes can only be switched between two complete BIST iterations.

In the BIST iterations using Fibonacci LFSR, the last right shifting operation prepares the PG of the next iteration. In the case of switching the PG mode, the right shifting operation is redundant, which introduces extra latency and power consumption. For the switch from the Fibonacci LFSR mode to other PG modes, the latency cost is two clock cycles. One cycle is for the redundant right shifting operation, and the other cycle resets the state to IDLE. The simulation result shows that the energy consumption of the redundant right shifting operation is 10.74 fJ/bit. For the switch between the other two PG modes, the latency cost is only one clock cycle.

On the other hand, since the workflows of PG and RA are merged to form a complete BIST iteration, extra type-A bitcells are required to store the test patterns and the signatures separately. For example, as shown in Fig. 5(c) and (g), the Fibonacci LFSR is implemented with only two type-A bitcells in each LXM. To store the signature in MISR, an extra type-A bitcell has to be used, which brings the area overhead of six transistors. Consequently, the area overhead of the proposed BIST scheme increases about 8.1% for the switch from the Fibonacci LFSR or CA-based PRPG mode to the MISR mode. For the switch from the WLFSR mode to the MISR mode, the ratio of area overhead increase depends on the weight factor f of WLFSR. In the case of $f = 63$, the total area overhead increases about 5.8%.

D. Comparisons

This section compares the proposed scheme with the indirect test method, the scan-based test approaches in [29], and the conventional LBIST schemes BEST and CBILBO.

The cycles counts consumed by the indirect and the scan-chain-based test approaches are linear with the number of PIs of the CUT. For an evaluation circuit with *pi* PIs and po POs, the indirect and the scan-chain-based approaches need *pi* cycles to write and scan in a test pattern, respectively. For the indirect approach, assume a standard po-stage scan

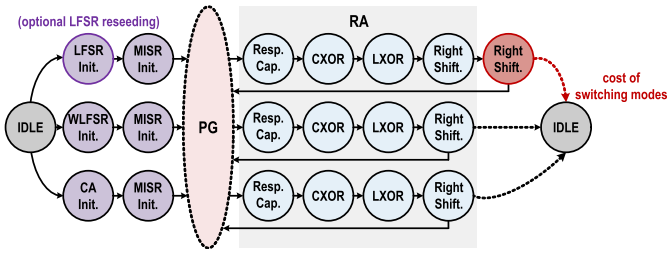


Fig. 13. State transition graph of the proposed BIST scheme.

chain is deployed at the primary outputs (POs) to capture the response. For the scan-chain-based approach, both the PIs and the POs of the CUT are connected with the same scan chain. The scan chain in both approaches above takes three more cycles to capture and hold the response [32]. The indirect and the scan-chain-based approaches need *po* cycles to scan out the response, respectively. As a result, both approaches require *pi*+*po*+3 cycles to finish a test iteration. The cycle counts of a single test iteration are unrelated to the external PG techniques.

The cycle counts of the conventional LBIST schemes depend on the applied pattern generator. If the pattern generator is an LFSR or CA, the LBIST schemes only consume one cycle to generate a new pattern. If the pattern generator is WLFSR, the cycle count depends on the number of PIs of the CUT. The conventional WLFSR generates the weighted patterns in serial. After one bit of weighted pattern is generated, all the *pi*-bit data stored in WLFSR must be shifted out before the next bit is generated, so the LBIST schemes consume pi^2 cycles to generate an *pi*-bit weighted pattern. The conventional MISR consumes one cycle to compact each response. Consequently, to implement a BIST iteration, the LBIST schemes consume two cycles with LFSR or CA, and consume pi^2 +1 cycles with WLFSR.

The cycle counts of single test iteration are compared in Fig. 11. With the Fibonacci LFSR mode, the performance of the proposed BIST scheme is far better than that of the scan-chain-based/indirect methods, and is comparable with that of BEST/CBILBO. With the WLFSR mode, the proposed BIST scheme has superior performance to the counterparts. For the CUT with 128 PIs, 128 POs, and a weight factor of 63, the proposed BIST scheme has 15.24× and 963.82× higher performance than the scan-chain-based/indirect methods and BEST/CBILBO schemes, respectively. With the CA-based PRPG mode, the performance of the proposed BIST scheme is superior to that of scan-chain-based/indirect methods, whereas lower than that of BEST/CBILBO.

Table III compares the area overhead of the proposed BIST scheme with that of the counterparts. The only area overhead of the indirect and scan-chain-based approaches is the scan chain, because the PG and RA are implemented externally. For the fair comparison, multiple pattern generators, including Fibonacci LFSR, WLFSR, and CA-based PRPG circuits are integrated into BEST and CBILBO. The outputs of different generators are selected by the 3-1 multiplexers (MUX). The weight factor of the WLFSRs is set as 63. Table III shows that the area overhead of the proposed BIST scheme is larger than that of the scan-chain-based/indirect approaches, whereas it is smaller than that of BEST and CBILBO. For the case of *pi*

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TABLE IV
CEM OF DIFFERENT TEST SOLUTIONS

pi	weight factor=63				
	Indirect	Scan-chain	BEST	CBILBO	**Proposed**
8	10.5263	5.2631	2.1274	1.9623	**5.1610**
9	10.7143	5.3571	2.0140	1.8528	**5.7216**
16	11.4286	5.7143	1.1695	1.0810	**8.9810**
24	11.7647	5.8823	0.6053	0.7387	**11.6831**
25	11.7925	5.8962	0.4074	0.7201	**11.9659**
32	11.9403	5.9701	0.6053	0.5601	**13.6912**
48	12.1212	6.0606	0.4074	0.3771	**16.4743**
64	12.2137	6.1069	0.3069	0.2841	**18.3112**
80	12.2699	6.1350	0.2461	0.2279	**19.6141**
96	12.3077	6.1538	0.2461	0.2279	**20.5862**
112	12.3348	6.1674	0.1763	0.1632	**21.3394**
128	12.3552	6.1776	0.1544	0.1430	**21.9401**

=128, the proposed BIST scheme has 27.0% and 32.4% less area overhead than the BEST and CBILBO, respectively.

A comprehensive efficiency metric (CEM), which harmonizes the performance, the area efficiency, and the reconfigurability, is defined to evaluate the holistic quality of the test solutions. The definition of CEM is as follows:

$$\text{CEM} = \frac{\alpha}{pi^2} \cdot \frac{3}{\frac{1}{C_l T_l} + \frac{1}{C_w T_w} + \frac{1}{C_c T_c}} \tag{6}$$

where the pi is the number of PIs, the C_l, C_w, and C_c are the cycle counts to implement single test iteration with Fibonacci LFSR, WLFSR, and CA-based PRPG, respectively, and T_l, T_w, and T_c are the corresponding amount of extra transistors, respectively. The α is a scale factor which is set as 1000. The greater CEM value represents a better holistic quality of test solutions. Table IV compares that of different test solutions. The CEM of the proposed BIST scheme increases with pi. Comparing with the conventional LBIST schemes, the proposed BIST scheme has more than 2.4× and 142.0× CEM gain when pi is 8 and 128, respectively. The CEM of the proposed BIST scheme outperforms the CEM of the scan-chain-based method when $pi > 8$, and it outperforms the CEM of the indirect method when pi >24. The comparison results indicate that the proposed scheme makes a good balance among the area overhead, the test performance, and the reconfigurability.

The superiority of the CEM of the proposed BIST scheme thanks to the following factors. In the aspect of area overhead, the IMC bitcells are reused to replace the DFFs in conventional LBIST schemes, while the in-array AND, OR, and XOR operations are realized without additional logic gates. Moreover, the MUXs used for pattern selection are eliminated because the PG technique in the proposed scheme can be reconfigured by simply applying different control signals. In the aspect of test performance, the proposed BIST scheme generates patterns and analyzes the responses near the CUT, thereby avoids the time consumption of data transfer. Furthermore, the proposed scheme exploits the parallelism of IMC, which significantly benefits its CEM when pi is large.

TABLE V
NOTATIONS OF OPERATIONS IN THE TEST OF THE PROPOSED BIST SCHEME

Operations	Notations
regular read	*R*
regular write	*W*
horizontal read	*HR*
horizontal write (through DIN_CUT or DIN_CXM)	*HW* (*HWCUT*, *HWCXM*)
right shift from cell*a* column to cell*b* column	$RSH\{a, b\}$
horizontal pseudo-write and horizontal pseudo-right-shift with deactivated END signal	*PHWCUT*, *PHWCXM* $PRSH\{a, b\}$
chain XOR/$\overline{\text{XOR}}$ with inputs from cell*a* column*	$CXOR\{\ldots, a_i^o, \overline{a_j^o}, \ldots\}$ $\overline{CXOR}\{\ldots, a_i^o, \overline{a_j^o}, \ldots\}$
local XOR with inputs from cell*a* and cell*b* columns, the operands are p and q	$LXOR\{a^p, b^q\}$

* a_i^o denotes the input of the chain XOR/$\overline{\text{XOR}}$ operation, where o denotes the operand of the input, which can be "0" or "1". $\overline{a_i^o}$ means that the input is disabled, thereby the corresponding operand does not participate in the chain XOR/$\overline{\text{XOR}}$ operation.

V. DISCUSSION

A. Test Method of the Proposed BIST Scheme

The effective testing of the proposed BIST scheme is the prerequisite of its application in digital SRAM-IMC architectures. In this section, the functional test methods for the CXM stages, SDMs, and LXMs of the proposed BIST scheme are discussed, respectively. For the CXM stages, SDMs, and the NOR gates in the LXMs, the stuck-at faults (SAFs) are considered. For the customized SRAM bitcells in the LXMs, the static functional fault models (FFMs) in both the memory mode and the IMC mode are taken into account. The FFMs are denoted as $\langle S/F/R \rangle$ [39], where S, F, R denote the sensitization operations, the faulty value, and the read-out value of the faulty cell, respectively. Table V presents the notations of the operations used in the test.

To analyze the stuck-at 0 (SA0) and stuck-at 1 (SA1) faults in the CXM circuit, all the stem and the fan-out (FO) signals are presented in Fig. 14(a). We analyze the single stage of the CXM circuit, because the structures of all CXM stages are identical. According to the fault collapse and dominance theory [40], the SAFs in the CXM are collapsed, and the corresponding sensitization conditions are presented in Table VI.

All the stem and the FO signals in the SDM circuit is presented in Fig. 14(b). Table VII shows the collapsed SAFs in the SDM circuit and the corresponding sensitization conditions. Note that the SAFs on DIN_CUT port and DIN_CXM port are merged into the SAF set in CXM and CUT, respectively. The *HWCXM1* operation is realized by deactivating all $\overline{\text{ENT}}$ signals of CXM, and setting $\overline{\text{ENX}}$ as "1." Logic "1" is generated at the output of each CXM stage in this case, thereby logic "1" is written into all LXMs through the DIN_CXM ports of SDMs. Similarly, the *PHWCXM1* operation also deactivates the $\overline{\text{ENT}}$ signals and sets $\overline{\text{ENX}}$ as "1," whereas the END signal is not activated. Besides, the *PRSH* operation

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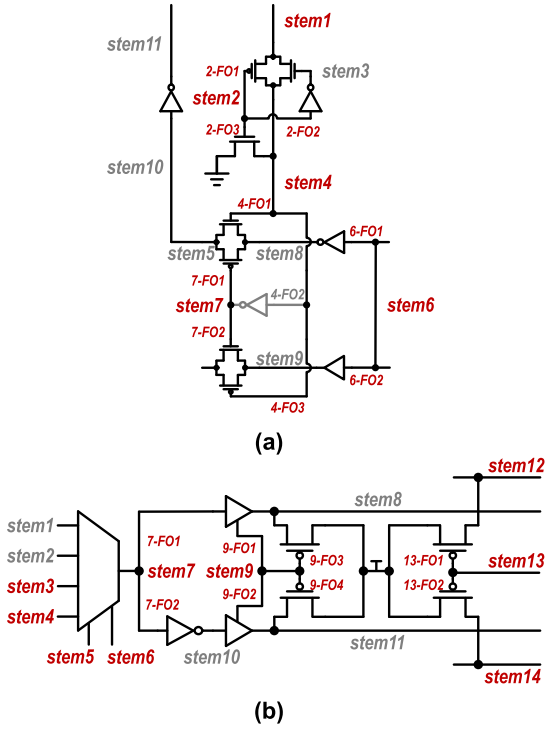


Fig. 14. Stem and FO signals in (a) one stage of the CXM circuit and (b) SDM circuit. The signals with collapsed SAFs are marked in red.

TABLE VI

COLLAPSED SAFS AND THE CORRESPONDING SENSITIZATION OPERATIONS FOR THE CXM CIRCUIT

Sensitization Operation*	Covered Collapsed SAFs
$CXOR\{a_t^1\}\cup\{\overline{a_i^0}\|i\in\{0, 1, \ldots, n\text{-}1\}, i\neq t\}$	stem1-SA0; stem2-SA1; 2-FO1-SA1; 2-FO3-SA1; stem4-SA0; 4-FO3-SA0; stem6-SA0; 6-FO1-SA0; 7-FO1-SA1; stem8-SA1
$CXOR\{a_t^0\}\cup\{\overline{a_i^0}\|i\in\{0, 1, \ldots, n\text{-}1\}, i\neq t\}$	stem1-SA1; 2-FO2-SA1; stem4-SA1; 4-FO3-SA1; 6-FO2-SA0; 7-FO1-SA0
$\overline{CXOR}\{a_t^1\}\cup\{\overline{a_i^0}\|i\in\{0, 1, \ldots, n\text{-}1\}, i\neq t\}$	4-FO1-SA0; stem6-SA1; 6-FO1-SA1; 7-FO2-SA1; stem8-SA0
$\overline{CXOR}\{a_t^0\}\cup\{\overline{a_i^0}\|i\in\{0, 1, \ldots, n\text{-}1\}, i\neq t\}$	4-FO1-SA1; 6-FO2-SA1; 7-FO2-SA0
$CXOR\{\overline{a_t^1}\}\cup\{\overline{a_i^0}\|i\in\{0, 1, \ldots, n\text{-}1\}, i\neq t\}$	stem2-SA0; 2-FO1-SA0; 2-FO3-SA0
$CXOR\{\overline{a_t^0}\}\cup\{\overline{a_i^1}\|i\in\{0, 1, \ldots, n\text{-}1\}, i\neq t\}$	2-FO2-SA0

* a_t denotes the input of the under-test stage in CXM circuit.

is also implemented by carrying out the *RSH* operation without activating END. By the *PHWCXM* and *PRSH* operations, the SA1 faults on the END signal, i.e., stem9 in Fig. 14(b), and its fanouts can be sensitized.

The fault behaviors of the type-A bitcells, the type-B bitcell, and the NOR gate in LXM are discussed, respectively. For the customized type-A and type-B bitcells, only the static faults related to the horizontal read and write ports are under consideration, because the faults in the rest parts are detected by the test of the bitcell array. All the bitcells support four *HW* functions (1*HW*0, 0*HW*0, 0*HW*1, 1*HW*1), and two *HR* functions (0*HR*0, 1*HR*1). Furthermore, the *LXOR* functions are supported by any two of the type-A bitcells. The *HW* functions of type-A bitcells are tested through the *RSH* and *HWCXM* operations. On the other hand, the *HR* functions of type-A bitcells and the *HW* functions of type-B bitcell are tested by the *LXOR* operations, for the *LXOR* operations are based on the simultaneous *HR* of type-A bitcells and the *HW* of type-B bitcell. The SAFs on the NOR gate are also covered by the *LXOR* operations. Accordingly, the collapsed faults and the corresponding sensitization conditions are analyzed and presented in Table VIII.

TABLE VII

COLLAPSED SAFS AND THE CORRESPONDING SENSITIZATION OPERATIONS FOR THE SDM CIRCUIT

Sensitization Operation*	Covered Collapsed SAFs
$W0$ to cell$a$ column; $W1$ to cell$b$ column; $LXOR\{a^0, b^1\}$; $RSH\{4, c\}$	stem3-SA0
$W1$ to cell$a$ column; $W1$ to cell$b$ column; $LXOR\{a^1, b^1\}$; $RSH\{4, c\}$	stem3-SA1
$W1$ to cell$a$ column; $W0$ to cell$b$ column; $RSH\{a, b\}$	stem4-SA0
$W0$ to cell$a$ column; $W1$ to cell$b$ column; $W1$ to cell$c$ column; $RSH\{a, b\}$; $PRSH\{a, c\}$	stem4-SA1; stem5-SA0; stem6-SA1; stem7-SA1; 7-FO1-SA1; 7-FO2-SA1; stem9-SA0; 9-FO1-SA0; 9-FO1-SA1; 9-FO3-SA0; 9-FO3-SA1
$W0$ to cell$a$ column; $W0$ to cell$b$ column; $HWCXM1$ to cell$a$ column; $PHWCXM1$ to cell$b$	stem5-SA1; stem6-SA0; stem7-SA0; 7-FO1-SA0; 7-FO2-SA0; stem9-SA1; 9-FO2-SA0; 9-FO4-SA0
$W1$ to cell$a$ column; $W0$ to cell$b$ column; $HR1$ from cell$a$ column; $HR0$ from cell$b$ column	stem12-SA0; stem13-SA0; 13-FO1-SA0; 13-FO1-SA1; stem14-SA1
$W0$ to cell$a$ column; $W1$ to cell$b$ column; $HR0$ from cell$a$ column; $HR1$ from cell$b$ column	stem12-SA1; stem13-SA1; 13-FO2-SA0; 13-FO2-SA1; stem14-SA0

* cell$a$, cell$b$ and cell$c$ can be any three different type-A bitcells.

With the aforementioned sensitization conditions, a test algorithm of the proposed BIST scheme is illustrated as Algorithm 1. The test complexity of n-stage CXM is $18n$. For n SDMs and n LXMs, the test complexity is $8n+61$, because all operations except the regular write (W) are executed in parallel for all SDMs or LXMs. Consequently, the test algorithm for the proposed BIST scheme is with the complexity of $26n+61$.

B. Repair Method of the Proposed BIST Scheme

The repair method is required if some faults in the BIST scheme are detected. The repair method of the proposed BIST scheme is implemented by introducing redundant circuit modules.

First, in order to repair the faulty LXMs, the redundant columns of type-A and type-B bitcells are integrated into the LXMs. If there are faulty bitcells in LXMs, the corresponding bitcell columns are remapped to the redundant columns.

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TABLE VIII

COLLAPSED SAFS AND THE CORRESPONDING SENSITIZATION OPERATIONS FOR THE LXM CIRCUIT

Sensitization Operation*	Covered Static Memory FFMs and Collapsed SAFs
$W0$ to cell$a$ column; $W0$ to cell$b$ column; $RSH\{a, b\}$	$\langle 0HW0/1/\text{-}\rangle$ for cell$b$
$W0$ to cell$b$ column; $HWCXM1$ to cell$b$ column	$\langle 0HW1/0/\text{-}\rangle$ for cell$b$
$W1$ to cell$b$ column; $HWCXM1$ to cell$b$ column	$\langle 1HW1/0/\text{-}\rangle$ for cell$b$
$W0$ to cell$a$ column; $W1$ to cell$b$ column; $RSH\{a, b\}$	$\langle 1HW0/1/\text{-}\rangle$ for cell$b$
$W0$ to cell$a$ column; $W0$ to cell$b$ column; $LXOR\{a^0, b^0\}$	$\langle LXOR\{a^0, b^0\}/1/\text{-}\rangle$, $\langle 0HR0/0/1\rangle$, $\langle 0HR0/1/1\rangle$, $\langle 0HR0/1/0\rangle$ of cell$a$, cell$b$; $\langle 0HW1/0/\text{-}\rangle$, $\langle 1HW0/1/\text{-}\rangle$ of cell4; SA1 faults on NOR gate inputs
$W0$ to cell$a$ column; $W1$ to cell$b$ column; $LXOR\{a^0, b^1\}$	$\langle LXOR\{a^0, b^1\}/0/\text{-}\rangle$ of cell$a$, cell$b$; $\langle 0HW0/1/\text{-}\rangle$, $\langle 1HW1/0/\text{-}\rangle$ of cell4; SA0 faults on NOR gate inputs
$W1$ to cell$a$ column; $W1$ to cell$b$ column; $LXOR\{a^1, b^1\}$	$\langle LXOR\{a^1, b^1\}/1/\text{-}\rangle$, $\langle 1HR1/1/0\rangle$, $\langle 1HR1/0/0\rangle$, $\langle 1HR1/0/1\rangle$ of cell$a$, cell$b$;
$W1$ to cell$a$ column; $W0$ to cell$b$ column; $LXOR\{a^1, b^0\}$	$\langle LXOR\{a^1, b^0\}/0/\text{-}\rangle$ of cell$a$, cell$b$;
$HR0$	$\langle 0HR0/0/1\rangle$, $\langle 0HR0/1/1\rangle$, $\langle 0HR0/1/0\rangle$ of cell4;
$HR1$	$\langle 1HR1/1/0\rangle$, $\langle 1HR1/0/0\rangle$, $\langle 1HR1/0/1\rangle$ of cell4;

* cell$a$, cell$b$ can be any two of the type-A bitcells.

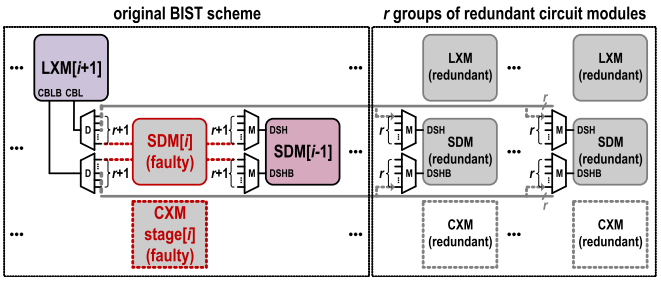


Fig. 15. Repair method of the faulty SDMs and the CXM stages in the proposed BIST scheme.

Second, in order to repair the faulty SDMs or CXM stages, the redundant SDMs and CXM stages are set. The redundant BIST circuit module groups, i.e., groups of single LXM, SDM, and CXM stages, are integrated with the redundant rows of the bitcell array. As presented in Fig. 15, to select the signals from r redundant groups, two demultiplexers (DEMUX) and two MUXs are connected to each LXM and SDM in the original BIST scheme, respectively. Each DEMUX has r+1 outputs, one of the outputs is fed to the right adjacent SDM, the rest r outputs are connected to the r redundant SDMs. Similarly, each MUX has r+1 inputs, one of the inputs receives the shift data from the left adjacent LXM, while the remaining r inputs collect the data outputs from the r redundant LXMs, respectively. The additional MUXs are also connected to the

Algorithm 1 Test Algorithm of the Proposed BIST Scheme

// test CXM circuit
$W0$ to cell1 column;
for t **in** range(n)**:**
  $W1$ to cell1[t];
  $CXOR\{1_t^1\}\cup\{\overline{1_i^0}|i\in\{0, 1, \ldots, n\text{-}1\}, i\neq t\}$;
  HR0 from cell1[n];
  $\overline{CXOR}\{1_t^1\}\cup\{\overline{1_i^0}|i\in\{0, 1, \ldots, n\text{-}1\}, i\neq t\}$;
  HR1 from cell1[n];
  $CXOR\{\overline{1_t^1}\}\cup\{\overline{1_i^0}|i\in\{0, 1, \ldots, n\text{-}1\}, i\neq t\}$;
  HR0 from cell1[n];
  $W0$ to cell1[t];
  $CXOR\{1_t^0\}\cup\{\overline{1_i^0}|i\in\{0, 1, \ldots, n\text{-}1\}, i\neq t\}$;
  HR0 from cell1[n];
  $\overline{CXOR}\{1_t^0\}\cup\{\overline{1_i^0}|i\in\{0, 1, \ldots, n\text{-}1\}, i\neq t\}$;
  HR1 from cell1[n];
$W1$ to cell1 column;
for t **in** range(n)**:**
  $W0$ to cell1[t];
  $CXOR\{\overline{1_t^0}\}\cup\{\overline{1_i^1}|i\in\{0, 1, \ldots, n\text{-}1\}, i\neq t\}$;
  HR0 from cell1[n];
  $W1$ to cell1[t];
// test SDM and LXM circuit
for (a, b, c) **in** {(1, 2, 3), (2, 3, 1), (3, 1, 2)}**:**
  $W1$ to cell$a$, cell$b$ columns;
  HWCXM1 to cell$b$ column;
  HR1 from cell$b$ column;
  $LXOR\{a^1, b^1\}$; RSH{4, c};
  HR0 from cell$c$ column;
  $W0$ to cell$a$ column;
  $LXOR\{a^0, b^1\}$; RSH{4, c};
  HR1 from cell$c$ column;
  RSH{a, b}; HR0 from cell$b$ column;
  $LXOR\{a^0, b^0\}$; HR0 from cell4 column;
  RSH{a, b}; HR0 from cell$b$ column;
  HWCXM1 to cell$a$ column;
  $LXOR\{a^1, b^0\}$; HR1 from cell4 column;
$W0$ to cell1, cell2, cell3 columns;
HWCXM1 to cell2 column;
PHWCXM1 to cell3 column;
HR0 from cell3 column; HR1 from cell2 column;
$W1$ to cell3 column;
RSH{1, 2}; $PRSH$\{1, 3\};
HR1 from cell3 column; HR0 from cell2 column;
RSH{3, 1}; HR1 from cell1 column;

redundant SDMs. Each of these MUXs is with r inputs, in which one input is connected to every SDM in the original BIST scheme, and r-1 inputs are connected to the outputs of LXMs in other r-1 redundant circuit module groups.

The repair method for faulty SDMs and CXM stages is briefly illustrated as follows. Assume the SDM[i] in the original BIST scheme is faulty, the entire group of circuit modules, i.e., LXM[i], SDM[i], and CXM stage[i], is remapped to a redundant circuit module group. The right shift operation is realized by the DEMUXs and the MUXs. The DEMUXs connected to LXM[i+1] reroute the shift data to the redundant SDM, and the SDM[i-1] receives the shift data from the redundant LXM through the MUXs.

Assume the CXM stage[i] is faulty, the initialization and the chain XOR operations cannot be implemented correctly, because the inversion path and the bypass path in the CXM are

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TABLE IX
REPAIR ABILITY OF THE PROPOSED REPAIR METHOD

# Redundant Resources		# Tap Coeff. "1"	
r		e	
Repair Strategies	**# Repairable SDMs**	**# Repairable Temp. SDM***	**# Repairable CXM Stages**
strategy 1	r	0	0
strategy 2	r-e-1	1	any

* If the SDM in the temporary LFSR/MISR stage is faulty, the chain XOR result cannot be written back to the bitcell array normally. As a result, the repair method of the SDM is consistent with that of the CXM stages.

cut off by the faulty CXM stage. To realize the initialization operation, "0" and "1" are written to the redundant LXMs, then broadcasted to the LXMs in the original BIST scheme, respectively. To realize the chain XOR operation, all the e LFSR/MISR stages with tap coefficient "1" and the temporary LFSR/MISR stage used for storing the chain XOR result are remapped to the redundant circuit module groups. The chain XOR operation is implemented by the redundant CXM stages, and the temporary LFSR/MISR stage is remapped to the leftmost redundant group to store the chain XOR result. If $r< e+1$, the faulty CXM stage is unrepairable. According to the analysis above, two repair strategies based on the repair method are developed. The repair ability of the repair strategies is presented in Table IX.

VI. CONCLUSION

This work proposes a reconfigurable BIST scheme for the evaluation circuits of digital SRAM-IMC architectures. The proposed BIST scheme outperforms the existing indirect and scan-chain-based test approaches and the conventional LBIST schemes in the following aspects.

The proposed BIST scheme achieves higher fault coverage. The PG and RA are implemented independently in the BIST process, thereby the exhaustive or pseudo-exhaustive tests are enabled. The reconfigurability of the feedback polynomials makes it possible to apply the LFSR reseeding, which enhances the fault coverage by applying deterministic patterns. Furthermore, the multiple kinds of pattern generators supported by the proposed BIST scheme also improves the coverage for RP-resistant faults.

The proposed BIST solution achieves a better balance between the test performance and area overhead, and has higher holistic quality. The cycle costs of the BIST workflows are significantly lower than that of indirect and scan-chain-based test approaches, because of the elimination of data transfer. By reusing the IMC bitcells and parallel IMC operations, the proposed BIST scheme has lower area overhead than the conventional LBIST schemes, meanwhile realizes comparable test performance.

The proposed scheme provides a uniform BIST solution for the evaluation circuits in different scales, because of its modular architecture and consistent workflow. For each additional PI of CUT, only one extra LXM, SDM, and CXM stage are required. The test time with the proposed scheme is constant in the LFSR and WLFSR modes, and it increases linearly in the CA-based PRPG mode with the PI counts of CUT.

The proposed BIST scheme is reliable. The Monte-Carlo simulation results indicate that the function of the proposed scheme is robust against PVT variations. Moreover, the proposed test and repair mechanisms enable the detection and repair of the faults in the proposed BIST scheme.

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